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The Development and Field Deployment of a Novel Ultra-Deep High-Pressure Mudline Suspension System for Jackup Drilling Operations in Kuwait's Offshore Shallow Waters

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Abstract

Kuwait's offshore drilling operations in the Arabian Gulf present unique technical challenges requiring innovative engineering solutions. This paper describes the development and successful field deployment of a novel ultra-deep high pressure mudline suspension system specifically engineered for jackup drilling operations in shallow water environments. This system addresses critical operational challenges including high H₂S concentrations, wellbore instability in unconsolidated shale formations, and ultra-deep drilling requirements under high pressure high temperature (HPHT) conditions.

The innovative mudline suspension system incorporates advanced metal-to-metal sealing technology, high-capacity serpentine collet designs, and conventional engineering methodologies to verify this new system through a comprehensive testing program. The casing program utilizes a complex multi-string design ranging from 38"/36" conductor to 7⁵/₈" production casing, with total cumulative suspended loads of 8,000,000 lbs.

Key innovations include hydraulic test tools with pressure energization, customized BOP adapters with retractable landing shoulders, and materials selection utilizing high strength, high nickel, and high chrome alloys for H₂S resistance. The system underwent rigorous qualification testing to API 17D standards and achieved successful field deployment, establishing new industry benchmarks for offshore HPHT drilling operations.

Introduction

The Arabian Gulf's offshore drilling environment presents a convergence of technical challenges that have historically limited deep reservoir access from shallow water platforms. As demonstrated in previous offshore developments, the combination of environmental constraints, geological complexities, and operational requirements necessitates innovative approaches to conventional drilling methodologies.

Offshore drilling operations in Kuwait's territorial waters are characterized by several critical factors that distinguish them from conventional shallow water drilling programs. High hydrogen sulfide concentrations

which often exceed industry standard thresholds, require specialized materials and safety protocols. Wellbore instability, resulting from unconsolidated formations consisting of unstable shale and sand-shale sequences, leads to frequent wellbore collapse events and stuck pipe incidents. Additionally, stringent environmental regulations governing drilling waste management and disposal impose operational constraints requiring innovative waste handling solutions.

The primary technical objective was to develop a completion system capable of accessing Jurassic-aged ultra-deep reservoirs from jackup drilling platforms positioned in shallow water depths. This operational profile represents a significant departure from conventional deepwater drilling approaches, where floating drilling vessels such as drill ships or semi submersibles typically access ultra-deep formations from greater water depths with inherently different operational constraints.

Previous successful HPHT developments, including North Sea Central Graben projects^{1,2} and Gulf of Mexico ultra-deep programs, have demonstrated the feasibility of extreme condition drilling. However, these precedents utilized different platform configurations and operational environments, necessitating adaptation of some of these proven technologies to Kuwait's specific operational requirements.

The development program encompassed comprehensive system design, materials qualification, prototype testing, and field validation phases. The successful deployment represents the first commercial application of jackup-based ultra-deep HPHT drilling technology in the Arabian Gulf region in offshore Kuwaiti waters.

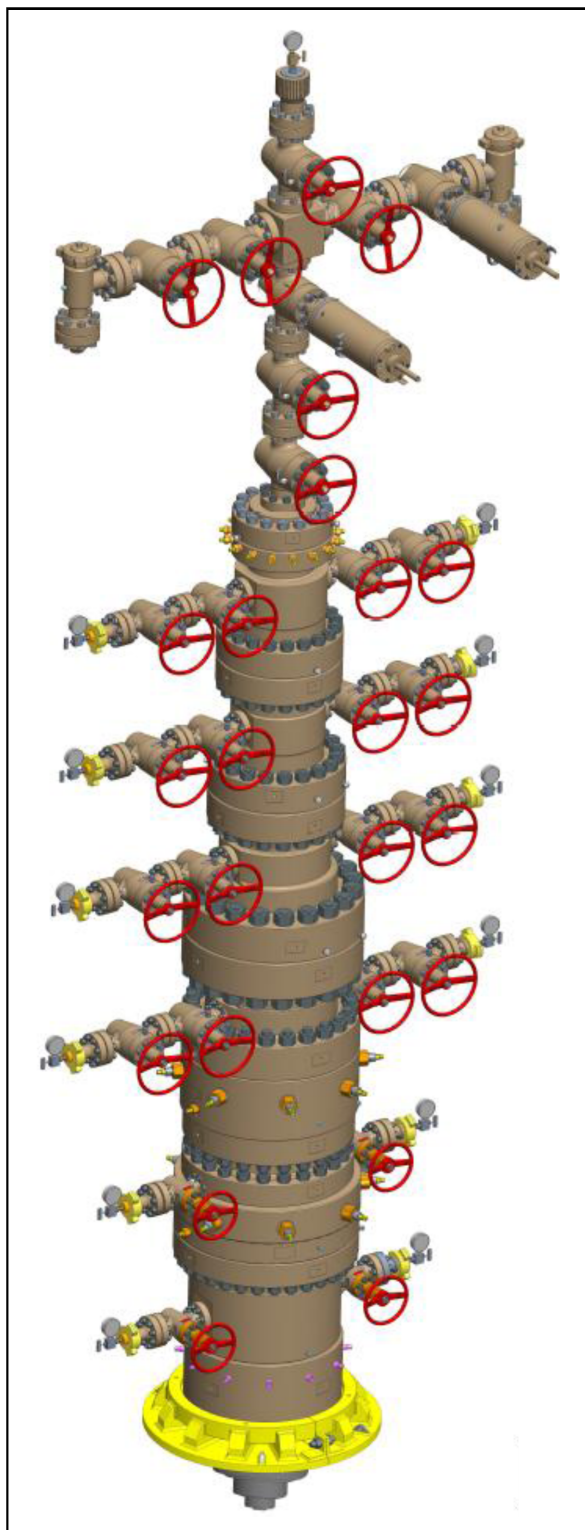


Figure 1—High Pressure Surface Wellhead System

Technical Background and Design Requirements

Operational Environment Definition

Offshore drilling in Kuwait's shallow waters required accommodation of unique operational parameters.

Shallow Water Jackup Configuration: Drilling operations utilized jackup rigs anchored in water depths typically ranging from 82 to 400 feet, to enable drilling through ultra deep formations with a stable platform foundation.

HPHT Reservoir Characteristics: The Jurassic reservoir is in a formation where pressures exceed 10,000 psi with temperatures approaching 350°F, requiring pressure control equipment and completion systems rated significantly above conventional shallow water applications.

Formation Instability Management: The presence of unconsolidated shale formations and sand-shale interbeds require specialized drilling fluid systems and wellbore support technologies to maintain hole integrity throughout extended drilling operations.

H₂S Environment: Due to the hazardous nature of hydrogen sulfide, its concentrations necessitate specialized metallurgy selection, including high strength, high nickel, and high chrome alloys, along with secondary barriers to prevent personnel from exposure.

Design Evolution and Development Program

The system development followed a two-phase structured progression from initial concept through field deployment. In phase one, the initial design efforts focused on developing mudline suspension capabilities for Cretaceous zone access with 10,000 psi working pressure equipment. The original casing program incorporated 24" × 18⁵/₈" × 13⁵/₈" × 10³/₄" × 7⁵/₈" × 4-1/2" tubing configurations, providing baseline experience for subsequent ultra-deep applications.

Following successful offshore campaigns in deploying the mudline system for Cretaceous wells, the development program progressed into the second phase venturing into the deeper and higher-pressure and more challenging Jurassic formation. This expansion necessitated comprehensive re-evaluation of equipment specifications, materials selection, and operational procedures to accommodate increased depth, pressure, and temperature parameters whilst working closely with operators to arrive at the optimum system.

The Jurassic development took multiple iterative phases involving conceptual design validation, establishing parameters and matching the mudline and surface wellhead equipment to complement the drilling feasibility assessment. This process repeated multiple times until the optimum solution was achieved.

The chronological order of the development program is shown below:

Table 1—Chronological Order of Jurassic Offshore MLL-E Mudline and Surface Wellhead Development

Date	9 Oct 2015	14 Apr 2016	23 Jun 2016	24 Nov 2016	24 Nov 2016	30 Dec 2016	15 Aug 2017
Name	Jurassic/Jurassic Alt Plus	Jurassic Deep 1	Jurassic Deep 2	Jurassic Deep 3	Jurassic Deep 4	Jurassic Deep 5	Jurassic New Onshore
Casing Program	42 × 30 × 24 × 18 5/8 × 13 5/8 × 10 3/4 × 7 5/8 15M	42 × 26 × 20 × 16 × 11 3/4 × 7 5/8 15M	42 × 24 × 18 5/8 × 13 5/8 × 10 3/4 × 7 5/8 15M	36 × 26 × 22 × 13 3/8 × 7 5/8 15M	36 × 26 × 20/22 × 13 5/8/13 3/8 × 10 3/4/9 5/8 × 7 5/8/7 15M	36 × 28 × 22 × 16 × 13 3/8 × 9 5/8 × 7 15M	36 × 30 × 24 × 18 5/8 × 13 5/8 × 10 3/4 × 7 5/8 15M
Cumulative Casing Weight (lbs)	7,736,369	5,725,621	5,132,244	7,878,365	8,069,511	7,7227,043	7,803,525
Stack Up Iterations	5	3	3	1	1	6	2
Technical Write up iterations	8	6	3	1	1	1	1
Design Calculations Program	7	3	3	1	1	1	2

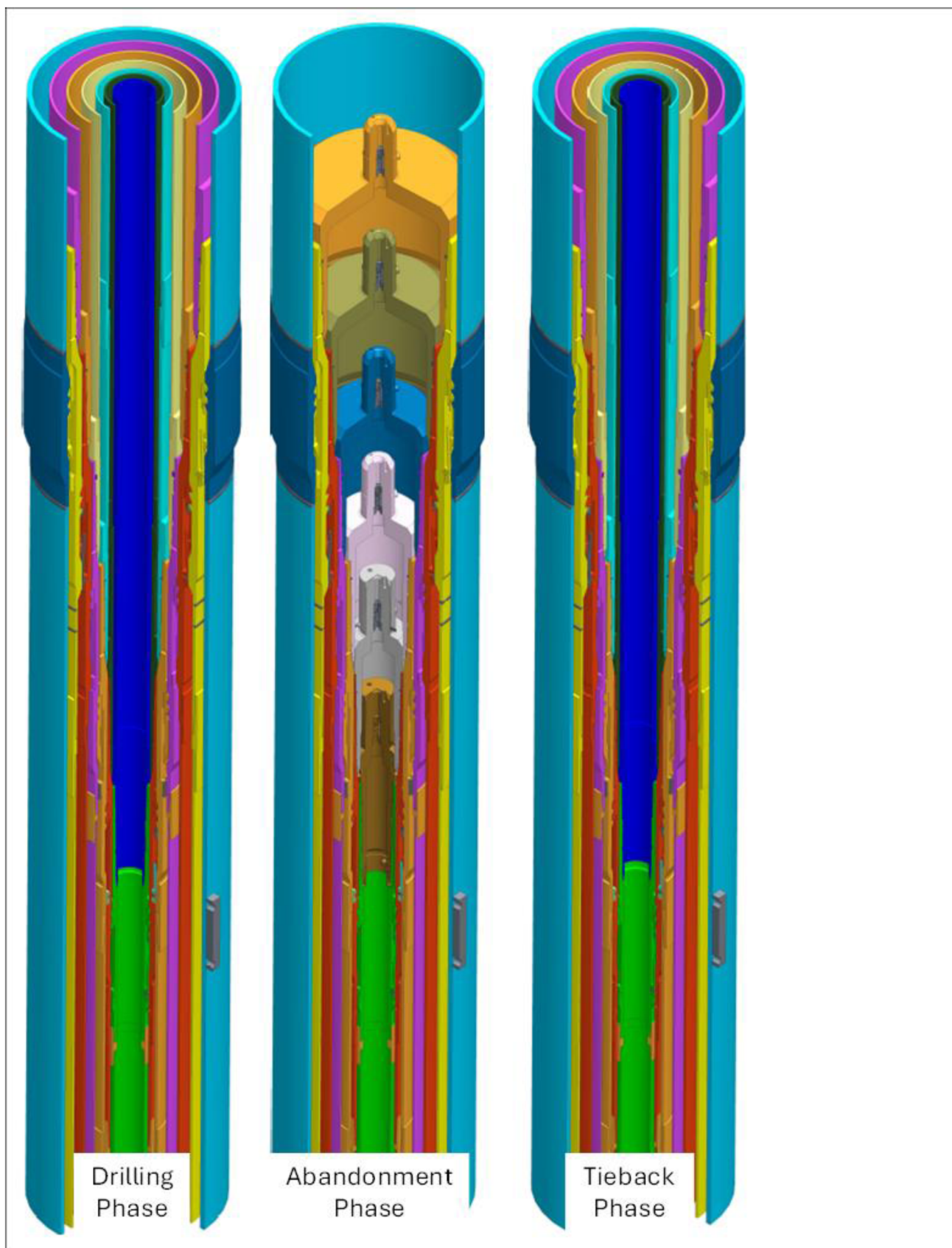


Figure 2—Mudline System in Drilling Phase, Abandonment Phase and Tieback Phase

Equipment Design and Component Development

Mudline Suspension System Architecture

The mudline suspension system provides the critical interface between the jackup drilling platform and the ultra-deep wellbore. The system design prioritized maximizing load capacity within extremely narrow geometric clearances imposed by multi-string casing programs and platform operational requirements.

Conservative Load Design Philosophy: All structural calculations utilized worst-case loading scenarios without considering buoyancy effects present in marine environments. Only air weight is considered in the calculations. This conservative approach provides additional safety margins and ensures reliable operation under all anticipated operational conditions.

Advanced Materials and Metallurgy

The extreme operational environment necessitated comprehensive materials evaluation and selection programs. High strength, high nickel, and high chrome materials were specified to provide adequate mechanical properties for HPHT conditions while resisting corrosion from high H₂S concentrations.

Materials selection criteria included: i) Minimum yield strength requirements for extreme loading conditions, ii) Corrosion resistance in high H₂S environments, iii) Mechanical strength stability through operating temperature range, and iv) Fabrication compatibility with complex geometries.

Finite Element Analysis Optimization: Comprehensive finite element analysis programs were employed to optimize metal-to-metal sealing interfaces, stress concentration areas, and load distribution patterns to achieve the desired contact stress. This analysis enabled refinement of critical dimensions and geometric features to ensure optimal performance under extreme conditions.

Complex Casing Program Implementation

Multi-String Casing Design

Access to ultra-deep Jurassic formations required implementation of a sophisticated multi-string casing program. The final configuration incorporates 38"/36" × 30" × 24" × 18⁵/₈" × 13⁵/₈" × 10³/₄" × 7⁵/₈" casing strings, with each section engineered for conventional drilling and installation while accommodating extreme operational parameters.

38"/36" Conductor System: The large-diameter conductor provides structural foundation and primary well control during initial drilling phases. Installation utilizes 49" diverter system with 37" packer isolation devices to swallow the 38"/36" pipe extending from the Landing Ring for well control and pressure management. A slim 38.50" Landing Ring is furnished with a double landing profile for landing a high-capacity serpentine collet style 30" Mudline hanger with a suspended capacity of 8,000,000lbs of casing load. The contoured profile of the serpentine collet is purposely designed to latch and land out on the landing profile only.

30" Surface Casing: The intermediate surface casing that is installed through 49" diverter system provides formation isolation and wellbore stability for subsequent drilling operations. The 30" MLL-E Mudline hanger lands onto the double landing profile inside the 38"/36" Landing Ring with a high-capacity serpentine collet that suspends a cumulative casing load of 8,000,000lbs. This hanger is furnished with dual top threads: left-hand thread for running tools and right-hand thread for tieback tools. This mudline system also utilizes pressure enhanced metal to metal sealing on the primary interfaces between the hanger/running tools, hanger/abandonment cap and hanger/tieback tools. After running and cementing of the 30" String with mudline system, the top of the 30" casing is brought back to surface where the 30" casing is cut and dressed for installation of a 30" Casing Head utilizing a non-welded fast make up connection to the 30" casing.

To meet the pressure requirements of drilling with 3,000psi pressure containment, a special drilling adapter is landed over the 30" 3M Casing Head spool. This special 30" 3M × 26³/₄" 3M drilling adapter is

equipped with retractable landing shoulders and pressure energized sealing system that permits full bore 26 $\frac{3}{4}$ " access whilst also providing the higher-pressure drilling at 3,000psi which would not have been possible otherwise as there is no 30" BOP or 30" annular rated to 3,000psi. Customized pressure energized test plugs are run into the 26 $\frac{3}{4}$ " flush bore to seal and isolate the bore to allow BOP pressure testing before running the 24" casing section.

24" Intermediate Casing: This section is run through the 26 $\frac{3}{4}$ " 3M BOP. The 24" MLL-E lands onto the double landing profile inside the 30" MLL-E Hanger with a high-capacity serpentine collet. This hanger is furnished with dual top threads: left-hand thread for running tools and right-hand threads for tieback tools. This mudline system also utilizes metal to metal sealing on the primary interfaces between the hanger/running tools, hanger/abandonment cap and hanger/tieback tools. After running and cementing the 24" String with mudline system, the top of the 24" casing is brought back to surface where the 24" casing is cut and dressed for installation of 30" $\times$ 24" Manual Slip Hanger inside the 30" Casing Head Housing after which the 30" $\times$ 26 $\frac{3}{4}$ " Casing Spool is installed above.

To meet the pressure requirements of drilling with 10,000psi protection, a special drilling adapter is landed over the 30" 3M $\times$ 26 $\frac{3}{4}$ " 5M Casing Spool. This special 26 $\frac{3}{4}$ " 5M $\times$ 21 $\frac{1}{4}$ " 10M drilling adapter is equipped with retractable landing shoulders and pressure-energized sealing system that permits full bore 21 $\frac{1}{4}$ " access whilst also providing the higher-pressure drilling at 10,000psi which would not have been possible otherwise as there is no 26 $\frac{3}{4}$ " BOP or 26 $\frac{3}{4}$ " annular rated to 10,000psi. Customized pressure energized test plugs are run into the 21 $\frac{1}{4}$ " flush bore to seal and isolate the bore to allow BOP pressure testing before running the 18 $\frac{5}{8}$ " casing section.

18 $\frac{5}{8}$ " Production Casing: The 18 $\frac{5}{8}$ " OD casing is run through the 21 $\frac{1}{4}$ " 10M BOP. The 18 $\frac{5}{8}$ " MLL-E lands onto the double landing profile inside the 24" MLL-E Hanger with a high-capacity serpentine collet. This hanger is furnished with dual top threads: left-hand thread for running tools and right-hand threads for tieback tools. This mudline system also utilizes metal to metal sealing on the primary interfaces between the hanger/running tools, hanger/abandonment cap and hanger/tieback tools. After running and cementing the 18 $\frac{5}{8}$ " String with mudline system, the top of the 18 $\frac{5}{8}$ " casing is brought back to surface where the 18 $\frac{5}{8}$ " casing is cut and dressed for installation of 26 $\frac{3}{4}$ " $\times$ 18 $\frac{5}{8}$ " Manual Slip Hanger inside the 26 $\frac{3}{4}$ " Casing Spool after which the 18 $\frac{3}{4}$ " 10M $\times$ 18 $\frac{3}{4}$ " 15M Casing Spool with a 26 $\frac{3}{4}$ " 5M $\times$ 18 $\frac{3}{4}$ " 10M DSA is installed above.

Drilling operations conducted through 18 $\frac{3}{4}$ " 15M BOP systems with conventional test plugs and wear bushings.

13 $\frac{5}{8}$ " through 7 $\frac{5}{8}$ " Completion Strings: Final casing sections utilize 18 $\frac{3}{4}$ " 15M BOP systems providing standard industry interfaces while maintaining full pressure control capabilities.

Performance Testing and Qualification

Comprehensive Testing Protocol

All mudline system components underwent rigorous testing programs exceeding industry standard requirements as the mudline system was novel due to the being the first of its kind with full metal to metal sealing, cumulative string weight of 8,000,000 lbs with a maximum pressure rating of 15,000psi. The qualification program incorporated API 17D compliance testing along with supplementary testing protocols addressing specific operational requirements.



Figure 3—System Integration Testing of all sections of the Mudline equipment

Structural Integrity Testing: All components were subjected to proof testing of design loads with verification of structural integrity and dimensional stability. Mudline Hangers, Running Tools and Tieback Tools were subject to Load Test as well as pressure tests to RWP.

Pressure Integrity Verification: Hydrostatic testing and gas testing up to 15,000 psi pressure validated pressure containment capabilities under extreme conditions for the mudline hangers, running tools, tieback tools and abandonment caps.

Corrosion Resistance Validation: Extended exposure testing in H_2S environments validated materials selection and corrosion resistance properties.

Pre-Deployment Validation

Simulated Fit and Function Testing: Complete system assemblies underwent simulated installation and operational testing to verify compatibility and identify potential operational issues prior to offshore deployment.

Quality Assurance Protocols: Comprehensive quality control measures ensured consistent manufacturing standards and dimensional accuracy throughout component fabrication.

Performance Verification: All critical components demonstrated performance capabilities exceeding design requirements with appropriate safety margins for operational variations.

Field Deployment and Operational Results: As of the timing of writing this article, the critical sections mentioned in this article have been successfully completed.

Installation Sequence and Procedures: Field deployment incorporated lessons learned from previous offshore HPHT operations while adapting to specific Kuwait offshore requirements. Installation procedures were developed through collaboration between equipment manufacturers, drilling contractors, and operating companies to ensure optimal operational efficiency.

Pre-Installation Preparation: Comprehensive equipment inspection and testing protocols verified system readiness and identified any transportation or handling effects requiring correction.

Sequential Installation Process: Casing installation followed established procedures with enhanced monitoring and verification protocols to ensure proper system function at each stage.

Operational Monitoring: Continuous monitoring of system performance parameters provided real-time validation of design assumptions and operational procedures.

Performance Validation

Field operations successfully demonstrated system capabilities under actual operational conditions. Key performance metrics included: i) Successful installation and operation of all casing strings, ii) Pressure integrity maintained throughout operational envelope, iii) Load management systems performed within design parameters, and iv) Environmental compliance maintained throughout operations

Innovation Assessment and Industry Impact

Technological Advancement

The successful development and deployment provide several significant technological advances:

1. Jackup-Based Ultra-Deep Drilling: The project demonstrates feasibility of accessing ultra-deep HPHT reservoirs from jackup platforms, expanding operational capabilities for shallow water environments.
2. Advanced Materials Application: Successful application of high-strength materials in large-bore offshore components establishes precedents for future extreme environment applications.
3. Innovative Sealing Technology: Metal-to-metal sealing systems validated under extreme conditions provide enhanced reliability compared to conventional elastomeric systems.
4. System Integration Methodologies: Successful integration of complex multi-string casing programs with advanced well control systems.

Quality and Reliability Enhancement

Conservative Design Philosophy: Implementation of worst-case design scenarios with exclusion of favorable environmental effects such as considering only air weight for all calculations provides enhanced reliability and operational safety margins.

Comprehensive Testing Validation: Extensive testing programs exceeding industry standards ensure component reliability and operational safety under extreme conditions.

Conclusions

The successful development and field deployment of the ultra-deep high pressure mudline suspension system represents a significant advancement in offshore drilling technology for shallow water environments. The project successfully addressed unique technical challenges while establishing new industry benchmarks for jackup-based ultra-deep HPHT operations.

Conflict of interest

The authors declare no conflict of interest.

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Nomenclature

API-	American Petroleum Institute
BOP-	Blowout Preventer
DSA-	Double Studded Adapter
HPHT-	High Pressure High Temperature
H ₂ S-	Hydrogen Sulfide
MLL-E-	Mudline Enhanced Suspension Equipment
psi-	Pounds per Square Inch
PSL-	Product Specification Level
RWP-	Rated Working Pressure
TVD-	True Vertical Depth

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